

ANDROID DEVELOPER INTERNSHIP PROGRAM

Task 1

Task Title:

Market Research & Competitor Analysis for App Idea

Research Report: Eco-Cart

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Field: Sustainability & Business

(Android Mobile Application)

1. App Idea Selection

Selected Idea: Eco-Cart: A sustainability assistant for mindful shoppers powered by AI technology.

Problem Statement: “The Intention-Action Gap” exists, as by 2026 consumers are more environmentally conscious than ever before. However, 70% of customers want to purchase a product that is sustainable, but this is difficult for them to recognize when in a grocery aisle. Existing labels are sometimes ambiguous ("greenwashing") and it can be time-consuming and frustrating to research the carbon footprint and brand ethics on the ground.

The Solution: Eco-Cart is an Android App developed using Flutter which enables consumers to scan any product code and immediately get an "Eco-Score. Eco-Cart is unlike any other health app, instead it deals with the Planet Score (Carbon footprint, packaging recyclability, and brand labor ethics).

2. Competitor Research (Top 3 Apps)

Feature	App 1: Yuka	App 2: EWG Healthy Living	App 3: Klima
Primary Focus	Personal Health & Nutrition	Chemical Safety/Toxicity	Carbon Footprint Tracking
UI/UX Quality	Excellent. Very fast scanner, clean interface.	Average. Often feels "academic" and cluttered.	Premium. High-quality graphics and gamification.
User Ratings	4.6/5 stars	2.0/5 stars	Not provided
Downloads	10M+	1M+	10k+
Strengths	Massive database; instant color-coded feedback.	Deep scientific backing for ingredient safety.	Excellent data visualization of personal impact.
Weaknesses	Ignores environmental impact and labor ethics.	UI is slow; limited to health, not "eco" factors.	Manual entry; no barcode scanner for products.

3. User Problem Identification

Based on a review of current sustainability apps in 2026, users are facing three main frustrations:

Information Silos: Users need to use one app for health (Yuka) and another app for carbon tracking (Klima). No "All in One" ethical shopping tool is available.

Greenwashing Confusion: Labels with little or no meaning. Users want a third party, verified score they can rely on.

No Alternatives: If an app presents a product as being “Bad”, very little is offered as a “Good” alternative, and one that can be found on the same shelf.

4. Gap Analysis

Market research indicates that there is a huge gap between user intent and technology available. The Play Store is full of "Health" and "Carbon Tracking" apps, but there is an essential "Action Gap" for the average consumer.

What is missing in current apps?

The available solutions are, in general, either of two types, both of which leave the user standing at the grocery store checkout lane

Personal Health Silhouette (e.g., Yuka): These are great at letting you know if a product will impact your body (calories, additives), but not the planet. A product that is low in sugar may earn a "Perfect" rating for a plastic heavy, high carbon product.

The "Post-Purchase" Reflectors (e.g. Klima): Apps that help you to guess your carbon footprint after the damage is done. They need manual data entry and cannot assist you with making better-in-time decisions when holding a product.

The "Data Wall": Most sustainability apps offer "Raw Data" (grams of CO2, liters of water), something that is difficult for a consumer to visualize on a regular basis. Users aren't looking for more data they're looking for a clear verdict.

Unique Selling Point (USP): The "Better Pick" Suggestion Instead of just giving a "Bad Score" and stopping there, **Eco-Cart** immediately shows a list of "**Better Picks**" in the same category.

- **How it works:** If you scan a "Brand A" bottle of water that has a low score, the app shows a small popup saying: "Try Brand B! It uses 100% recycled plastic."
- **Why this is different:** It moves the user from "Problem" to "Solution" instantly. It's not just a scanner; it's a **shopping assistant**.

5. Final App Concept (Intern-Friendly Version)

App Name: Eco-Cart

Target Audience: Gen Z and Millennial "Conscious Shoppers" (Ages 18–40) who want to shop ethically but are short on time.

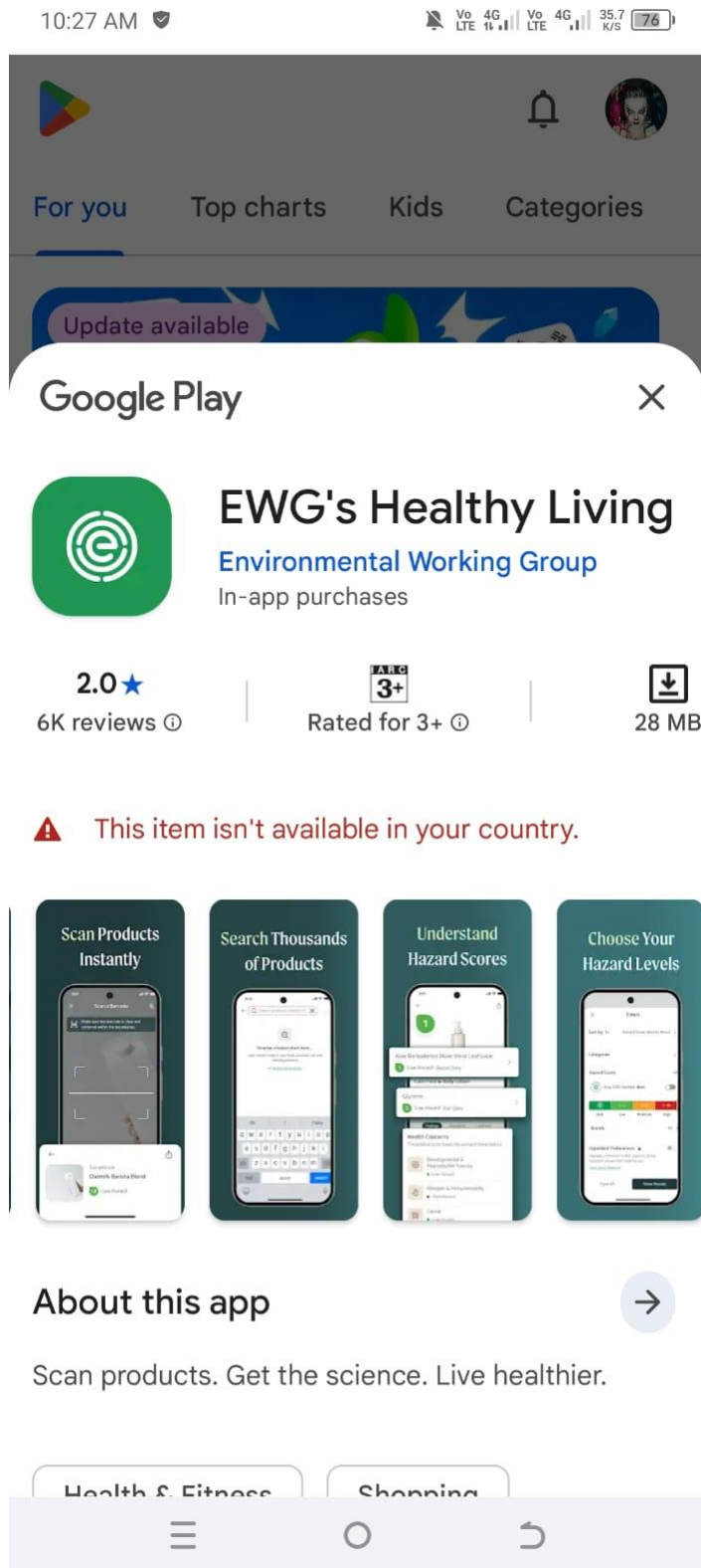
Core Features:

1. **AI Barcode Scanner:** Quick-scan feature using the mobile scanner package.
2. **Triple-Metric Score:** A simple "Traffic Light" system (Red, Yellow, Green) for Environment, Animals, and People.
3. **The "Better Pick" List:** A simple list that pops up if a product has a low score, suggesting 2–3 better alternatives.
4. **Local Impact Counter:** A dashboard that shows "Total Plastic Saved" or "Trees Saved" based on your better choices.
5. **Eco-Submit Form:** A simple screen where users can take a photo of a missing product and send it to you to add to the database.

Future Enhancements (No Blockchain!):

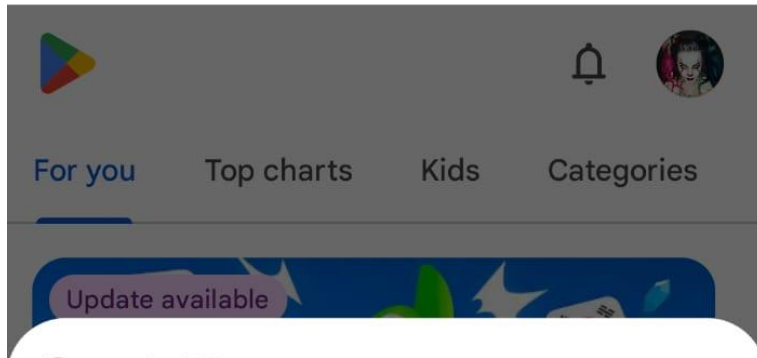
1. **Dark Mode & Low-Data Mode:** Since it's a "Green" app, adding a feature to reduce battery and data consumption to make the app itself more sustainable.
2. **Achievement Badges:** A simple gamification system where users unlock badges like "Plastic Warrior" or "Forest Friend" after 10 eco-friendly scans.

5. Screenshots



10:27 AM

Vo LTE 4G Vo LTE 4G 5.13 K/S 76



Google Play



2zero - Climathon

2zero - Gemeinsam das Klima schützen

Contains ads



24 MB

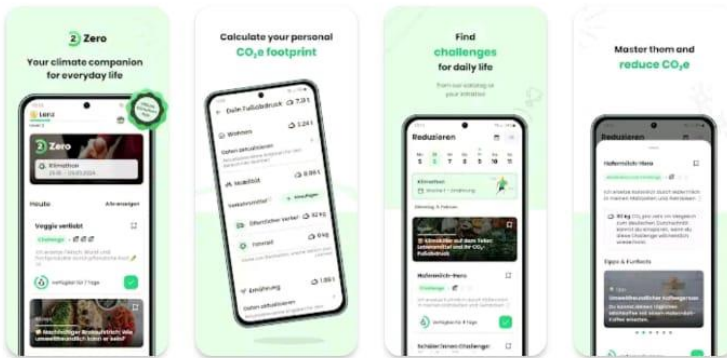


Rated for 3+ ⓘ

10K+

Downloads

Install



About this app

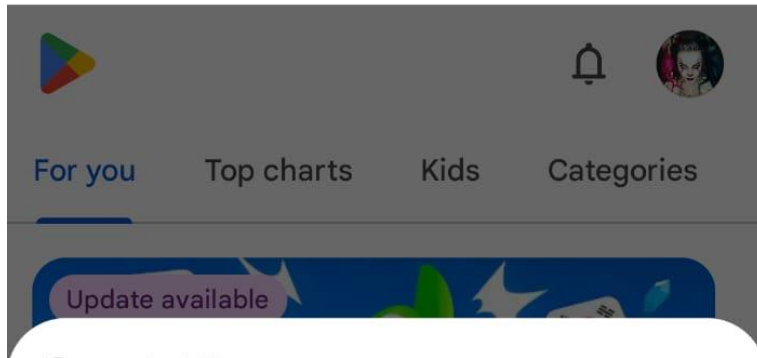


The Climathon app for your ecological footprint and environmental protection



10:26 AM

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Google Play



Yuka - Food & Cosmetic Scanner

Yuka App

In-app purchases



70 MB

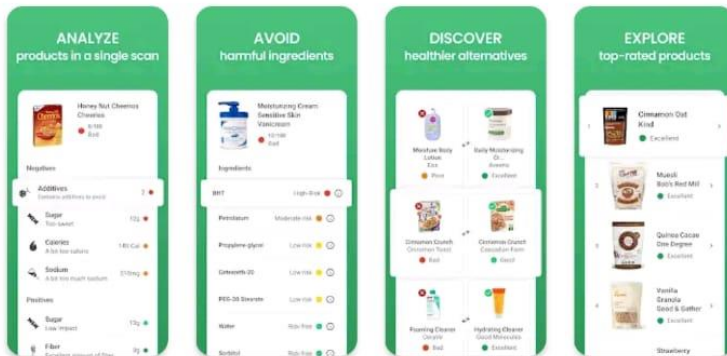


Rated for 3+ ⓘ

10M+

Downloads

⚠ This item isn't available in your country.



About this app



Scan food & cosmetic products and evaluate their impact on your health



6. Bonus: Extra Analysis

SWOT Analysis

1. **Strengths:** Solves a high-intent problem; high-quality Flutter UI/UX.
2. **Weaknesses:** Initial database of ethical scores will require heavy API research.
3. **Opportunities:** Partnering with eco-certified brands for "Verified" badges.
4. **Threats:** Larger apps like Yuka adding "Eco" features in the future.

Common User Complaints (from Competitors)

" Love what the app offers (EWG Healthy Living), but the glitches that everyone describes are frustrating. 1 - While searching, if you leave/ minimize the app, or even click on a product from list, your search is gone and you have to start over. 2- Won't make lists. I've used this app (and been annoyed by it) for years. I'd pay to have a version that actually worked."

Eco-Cart Response: We will prioritize **Performance Optimization** in Flutter to ensure the scanner never lags during a grocery trip.

Rough Wireframe Logic

1. **Home:** Large "Scan Now" button + User's current "Planet Rank."
2. **Scan Result:** A 0–100 circular progress bar (Green/Yellow/Red).
3. **Detail Page:** Three tabs: CO2 Impact, Packaging, Brand Ethics.